Module 4

Comparing Cloud Storage

Module Objectives

- Explore storage types and tiers.
- Explore storage performance and cost implications.

Lesson 4.1

Storage Resources and Technologies

Cloud+ CV0-004 Certification Exam Objective:

- 1.4 Compare and contrast storage resources and technologies.

Storage Resources and Technologies

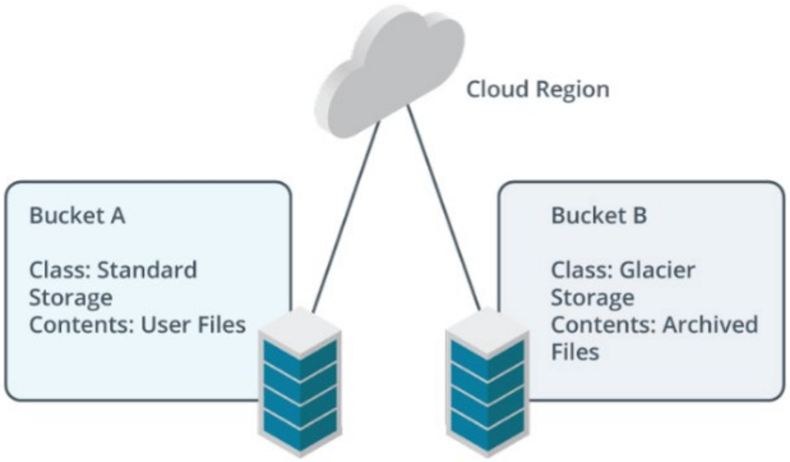
- Disk Types
- Storage Types
- Tiered Storage
- Migrate Data to the Cloud
- Performance Implications
- Storage as a Service
- Software-Defined Storage
- Cost Implications
- Hyperconverged Storage Resources

Disk Types

- Elastic Block Storage (EBS)
- Solid state drives (SSDs)
- Hard disk drives (HDDs)
- Hybrid disks
- Long-term storage

Storage Types

- Block storage
- File storage
- Object storage
- Buckets



Storage buckets with different classes and stored contents. (Images © 123RF.com)

Tiered Storage

- Storage tiers
  - Hot access tier
  - Warm
  - Cold
  - Archive
- Create VM Storage
  - Select the VM
  - Create a new disk or attach existing disk
  - Select disk-type, capacity, and performance setting

Migrate Data to the Cloud

	GCP	Azure	AWS
Direct Connection Options	• Direct Peering (public Internet) • Cloud Interconnect (avoids public Internet) • Storage Transfer Service (on-premises/cloud-to-cloud)	• ExpressRoute (avoids public Internet) • Migrate (all-in-one migration service)	• Cloud Data Migration • Direct Connect (avoids public Internet)
Physical device Options	• Transfer Appliance	• DataBox	• Snow Family

Performance Implications

- Input/output operations per second (IOPS)
- Throughput
- Latency
- Queue depth
- Analyzing performance

Storage as a Service

- Standalone STaaS examples
- Benefits of STaaS
- Potential drawbacks of STaaS

Software-Defined Storage (SDS)

- Compression
- Data deduplication
- Thick and thin storage provisioning
- Data replication

Cost Implications

- Egress fees
- Unnecessary data transfers/ synchronization
- Data requirements
- Data lifecycle
- Tiered storage
- Regional storage cost differences

Cost Implications: Manage User Quotas

- Storage filesystem
- Storage device
- Cloud filesystems
- Soft quotas
- Hard quotas

Hyperconverged Storage Resources

AWS M4 Instances

Instance	vCPU	Mem (GiB)	Storage	Dedicated EBS Bandwidth (Mbps)
m4.large	2	8	EBS-only	450
m4.xlarge	4	16	EBS-only	750

AWS D2 Instances

Instance	vCPU	Mem (GiB)	Storage
d2.xlarge	4	30.5	3x2TB HDD
d2.2xlarge	8	61	6x2TB HDD

Module Summary

- STaaS and SDS: Offer flexible cloud data placement options
- Benefits: Improved data proximity, availability, and recoverability
- Considerations: Data egress fees, tiered performance, and disk type choices
- Implications: Different from traditional on-premises storage